ESE December 2020_COMP_COA_2UCC303_Sem III_24 Dec_2.30 pm_Objective questions (20 marks)

* Required

1. Email address *

2. Enter your full name *

3. Enter your roll number *

4. Enter your division *

Mark only one oval.

- A
- B

Questions

5. How big is a four way set associative cache memory with a block size of 64 Bytes and containing 1024 sets? 1 point

Mark only one oval.

- Total size of cache memory is 262144 Bytes
- Total size of cache memory is 16384 Bytes
- Total size of cache memory is 65536 Bytes
- Not possible

6. A tiny bootstrap loader program is situated in ? 1 point

Mark only one oval.

- Hard disk
- ROM
- BIOS
- RAM

7. Cache and word addressable main memory system has the following specification:Cache block size = 16 words,Set size = 2 blocks,Number of sets = 128, Number of bits in a word = 32 bits,Size of main memory address = 21bits, What is the total cache size needed? 1 point

Mark only one oval.

- (16k + 140) bytes
- (16k + 320) bytes
- (60k + 300) bytes
- (16k + 300) bytes

8. A 5 stage pipeline has the stages IF, ID, OF, PO, WB (Assume that there are no separate data and instruction caches). For the program below, what is/are the hazard(s) possible? MOV R1,A MOV R2,A ADD R1,R2 MOV X,R1 — 1 point

Mark only one oval.

- Data Hazard
- Structural Hazard
- Control Hazard
- Both Data & Structural Hazard

9. Consider the following processor design characteristics. I. Register-to-register arithmetic operations only. II. Fixed-length instruction format. III. Hardwired control unit. Which of the characteristics above are used in the design of a RISC processor? — 1 point

Mark only one oval.

- I and II only
- II and III only
- I and III only
- I, II and III

10. Register renaming is done in pipelined processors ? — 1 point

Mark only one oval.

- As an alternative to register allocation at compile time
- For efficient access to function parameters and local variables
- To handle certain kinds of hazards
- As part of address translation

11. The main requirement for using single Bus structure is ___________ 1 point

Mark only one oval.

- Fast data transfers
- Cost effective connectivity and speed
- Cost effective connectivity and ease of attaching peripheral devices
- Multiple connectivity between devices

12. A CPU generally handles an interrupt by executing an interrupt service routine. 1 point

Mark only one oval.

- as soon as an interrupt is raised
- by checking the interrupt register at the end of fetch cycle.
- by checking the interrupt register after finishing the execution of the current instruction.
- by checking the interrupt register at fixed time intervals.

13. Delayed branching can help in the handling of control hazards, for all delayed conditional branch instructions, irrespective of whether the condition evaluates to true or false. 1 point

Mark only one oval.

- the instruction following the conditional branch instruction in memory is executed
- the first instruction in the fall through path is executed
- the first instruction in the taken path is executed
- the branch takes longer to execute than any other instruction

14. Consider the following sequence of micro-operations. MBR ← PC MAR ← X PC ← Y Memory ← MBR Which one of the following is a possible operation performed by this sequence? 1 point

Mark only one oval.

- Instruction fetch
- Operand fetch
- Conditional branch
- Initiation of interrupt service

15. The main advantage of multiple bus organisation over a single bus is _____ 1 point

Mark only one oval.

- Reduction in the number of cycles for execution
- Increase in size of the registers
- Better Connectivity
- Cost effectiveness

16. The representation of (309.1875) in double precision format is? 1 point

Mark only one oval.

- 0 10000111 001101010011000
- 0 10000000111 001101010011
- 0 10000001111 001101010011
- 0 01000000111 001101010011

17. A two way set associative cache consists of 32 words.Each block consists of 4 words. Size of the main memory is 256 words.What is number of bits in each of TAG,SET and WORD? 1 point

Mark only one oval.

- 4 2 2
- 6 2 2
- 3 3 2
- 2 2 4

18. Which addressing mode is indicated by the instruction ADD AX,[BX][SI+20]? 1 point

Mark only one oval.

- Based addressing mode
- Indexed addressing mode
- Based with displacement addressing mode
- Based indexed with displacement addressing mode

19. The control unit controls other units by generating __________? 1 point

Mark only one oval.

- Control signals
- Timing signals
- Transfer signals
- Command Signals

20. The time delay between two successive initiations of memory operation _______? 1 point

Mark only one oval.

- Memory access time
- Memory search time
- Memory cycle time
- Instruction delay

21. The technique where the controller is given complete access to main memory is __________? 1 point

Mark only one oval.

- Cycle stealing mode
- Memory stealing mode
- Transparent mode
- Burst mode

22. Consider the block sequence 2,3,2,1,5,2,4,5,3,2,5,2 with cache size of 3 blocks. What will be the final blocks in the cache when FIFO is used? 1 point

Mark only one oval.

- 5,3,2
- 2,3,5
- 3,5,2
- 2,5,3

23. A group of bits that tell the computer to perform a specific operation is called as?

1 point

Mark only one oval.

- Instruction code
- Micro-operation
- Register
- Accumulator

24. The ROM chips are mainly used to store _______?

1 point

Mark only one oval.

- System files
- Root directories
- Boot files
- Driver files